

CSIE 5452, Spring 2026: Homework 1

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1 Timing Analysis of the CAN Protocol: Part I (12pt)

1.1

Firstly, set $Q_0 = B_0 = C_1 = 30$, since no other task has higher priority than μ_0 , the terms in the summation is zero.

Hence, the worst-case response time $R_0 = Q_0 + C_0 = 30 + 10 = 40(\text{msec})$.

Iteration	LHS (Q_0)	B_0	RHS	Stop?
1	30	30	30	yes

1.2

Let $Q_1 = B_1 = C_1 = 30$ as the beginnning.

Iteration	LHS (Q_1)	B_1	j	$Q_1 + \tau$	T_j	$\lceil \frac{Q_1 + \tau}{T_j} \rceil$	C_j	RHS	Stop?
1	30	30	0	30.1	50	1	10	40	no
2	40	30	0	40.1	50	1	10	40	yes

Thus, the worst-case response time of μ_1 is $R_1 = Q_1 + C_1 = 40 + 30 = 70(\text{msec})$.

1.3

Let $Q_2 = B_2 = C_2 = 20$ as the beginnning. After 3 iterations, Q_2 converges. Hence, the worst case response

Iteration	LHS (Q_2)	$\lceil \frac{Q_2 + \tau}{T_0} \rceil C_0 + \lceil \frac{Q_2 + \tau}{T_1} \rceil C_1$	RHS	Stop?
1	20	40	60	no
2	60	50	70	no
3	70	50	70	yes

time $R_2 = Q_2 + C_2 = 70 + 20 = 90(\text{msec})$.

2 Timing Analysis of the CAN Protocol: Part II (36pt)

3 Timing Analysis of Preemptive Fixed-Priority Scheduling (16pt)

4 Timing Analysis of TDMA-Based Protocols (12pt)

5 MILP Linearization (12pt)

6 Signal Packing (12pt)